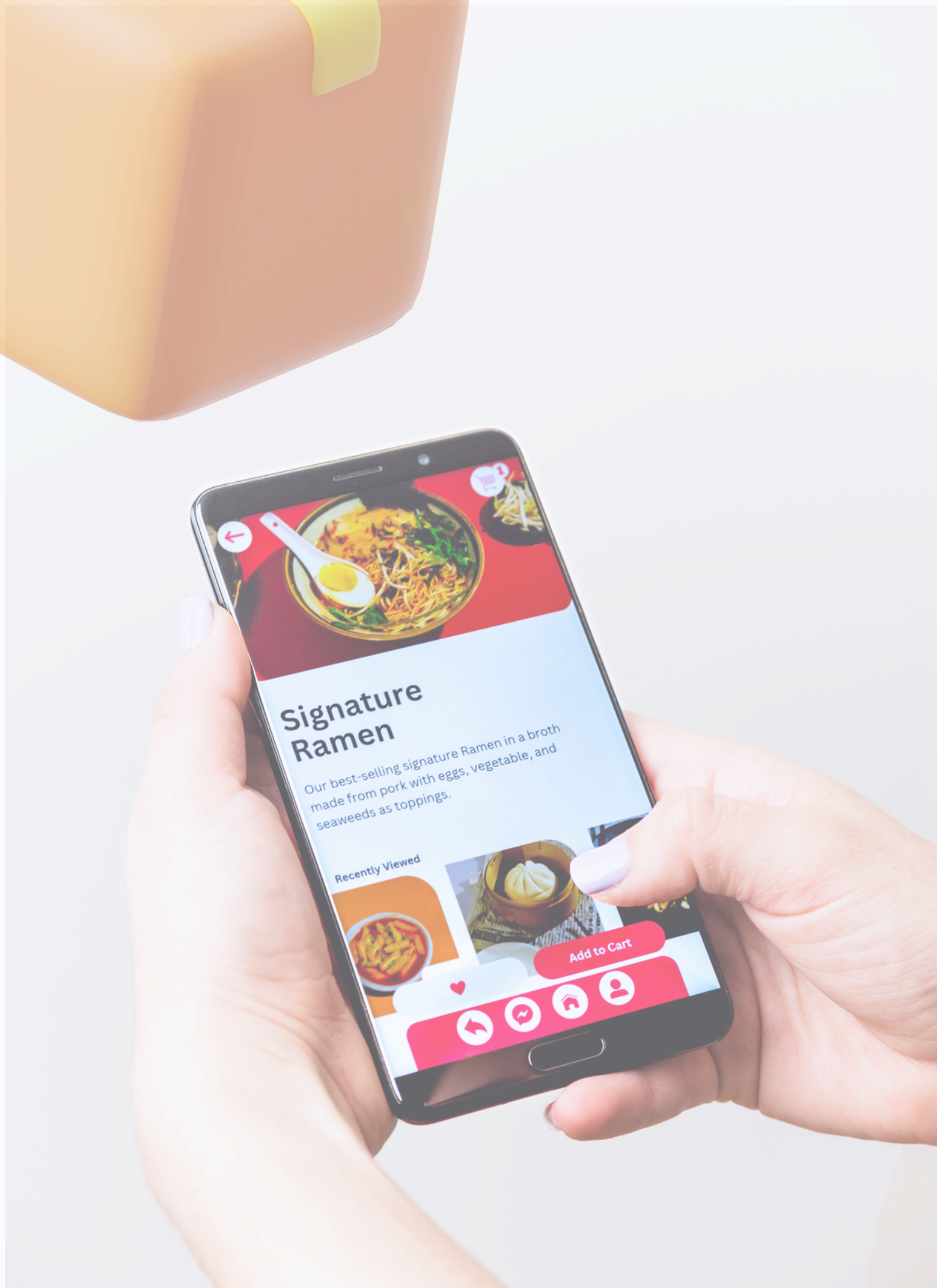




Zomato Delivery Operations – Business Intelligence Solution

Presented By:

Ouelhezi Molka Guizani Ons
Gaja Nour Ben Slama Ryhem



Introduction



- Zomato is a mid-sized food delivery company operating in multiple cities
- Partners with local restaurants; uses independent couriers (bike, scooter, car)
- **Key Operations:**
 - Order placement via mobile app
 - Restaurant order preparation
 - Courier assignment & delivery
 - Customer feedback & ratings
 - Collects large operational data for Business Intelligence analysis

The Business Problem



Delays and inconsistent delivery times frustrate customers.



Delivery services lack flexibility and real-time tracking.



Difficulty in accurate delivery time prediction

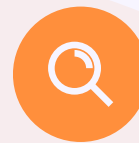


Service degradation during festivals and bad weather





The objectives



- Identify key factors affecting delivery time
- Evaluate delivery partner performance
- Improve delivery time prediction accuracy
- Enhance overall customer satisfaction and operational efficiency



Dataset & KPIs



Dataset Overview

- **Source:** Kaggle – Zomato Food Delivery Dataset
- **Description:** Historical operational delivery data including courier details, order timestamps, vehicle information, traffic density, weather conditions, and delivery time.
- **~46,000 deliveries across multiple cities**



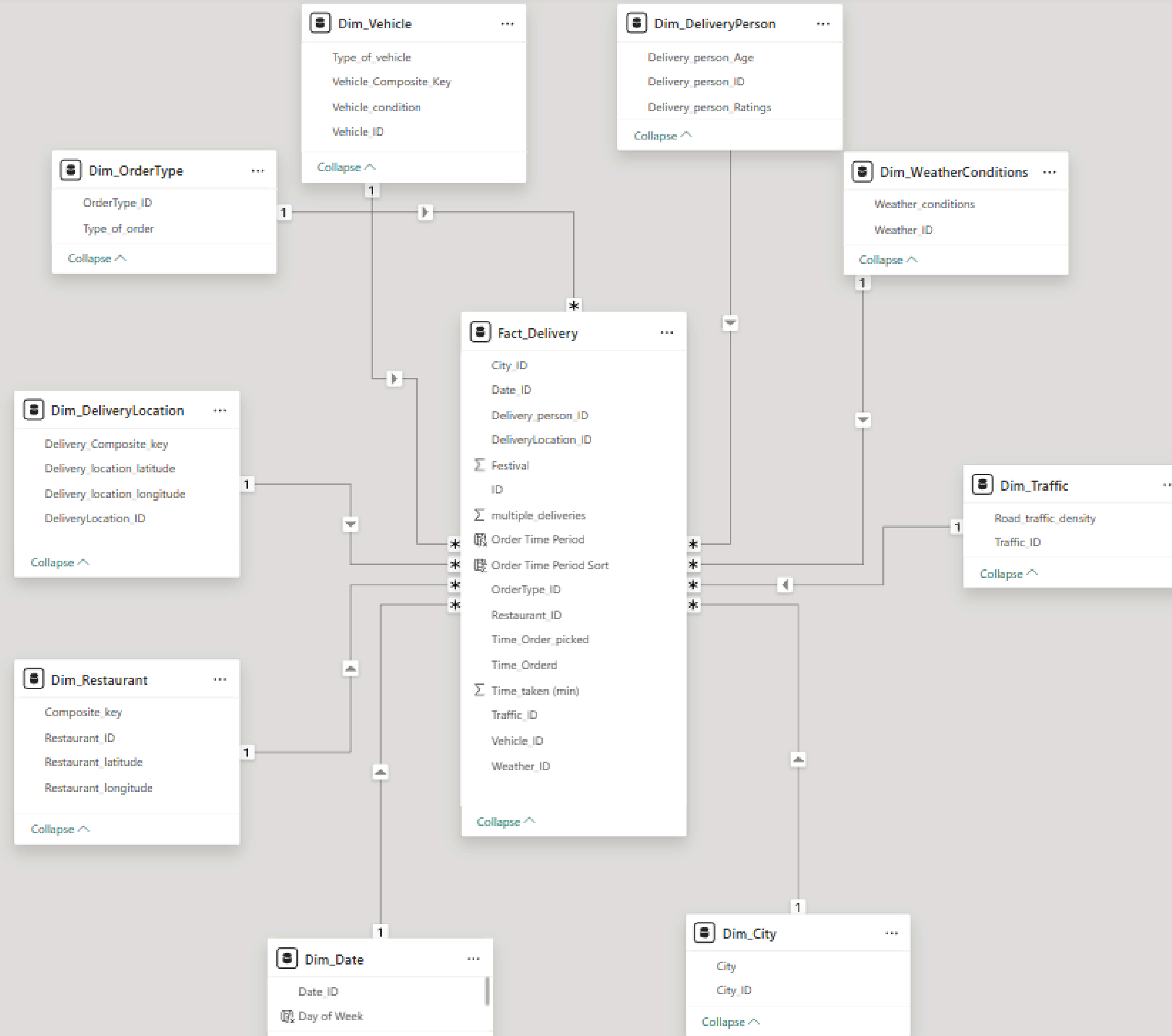
Key KPIs

- Average Delivery Time
- On-Time Delivery Rate
- Average Orders Delivered per Courier
- Average Delivery Time by Traffic Level
- Average Delivery Time during Festivals
- Average Delivery Time by Vehicle Type



Data Model

- **Dimensional Star Schema**
- Fact Table: Delivery fact
- Dimension Tables:
 - Date
 - City
 - Vehicle
 - Delivery Person
 - Traffic
 - Weather
 - Restaurant
 - Order type
 - Delivery location



ETL & Data Preparation

RAW DATA

- Zomato delivery operations dataset
- Format: CSV file
- ~46,000 records
- Data includes:
 - Order details
 - Delivery duration
 - Traffic level
 - Weather conditions
 - Vehicle type & condition
 - Courier information

Extract

- Load raw CSV file into Jupyter Notebook
- Read data using Pandas
- Inspect data structure and types
- Identify missing values and outliers

Transform

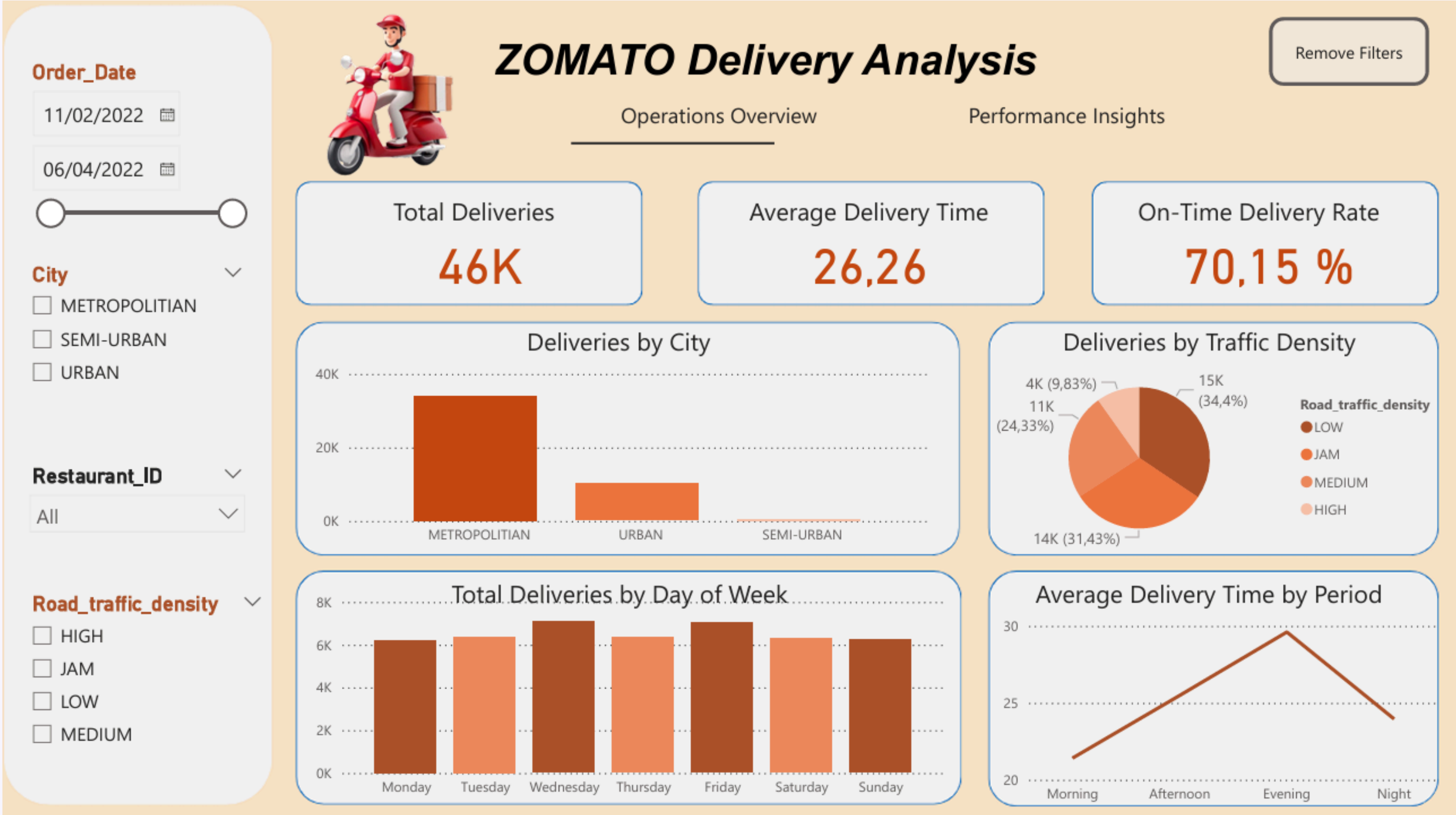
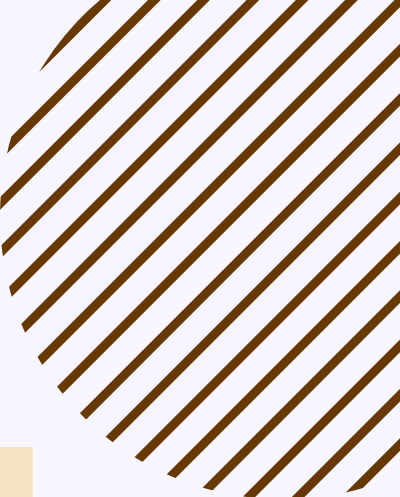
- Handle missing values
- Remove or cap extreme delivery times
- Standardize traffic and weather categories
- Convert data types (time, numeric fields)
- Handle Outliers
- Validate data quality and consistency

Load

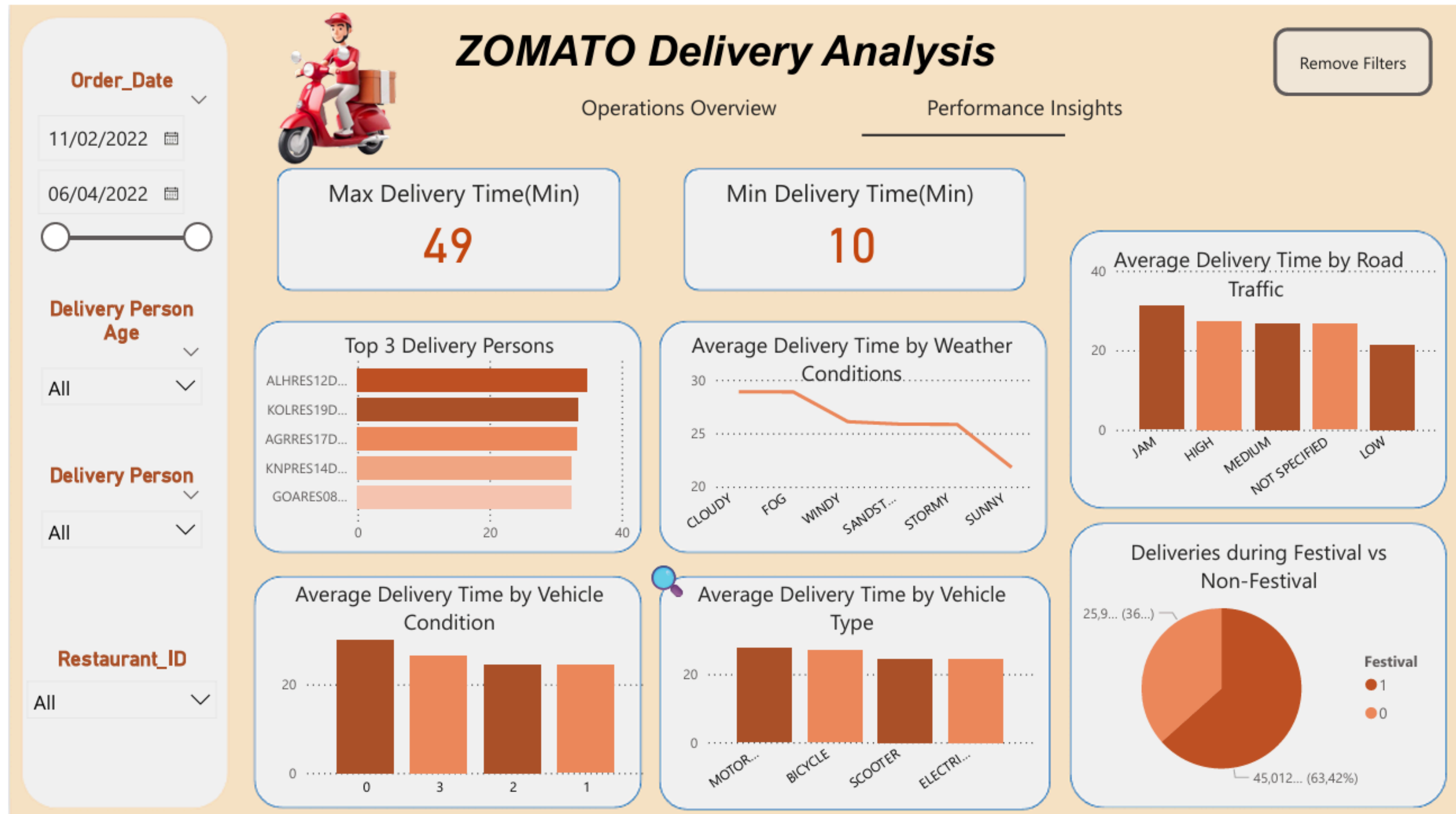
- Export cleaned dataset to CSV
- Load clean data into BI tool
- Build dimensional star schema
- Create KPIs and analytical measures



EXECUTIVE DASHBOARD



Deep Dive Dashboards





Delivery Performance for Selected Delivery Person

Delivery Person ID

AGRRES17DEL01

Delivery Person Age

36.0

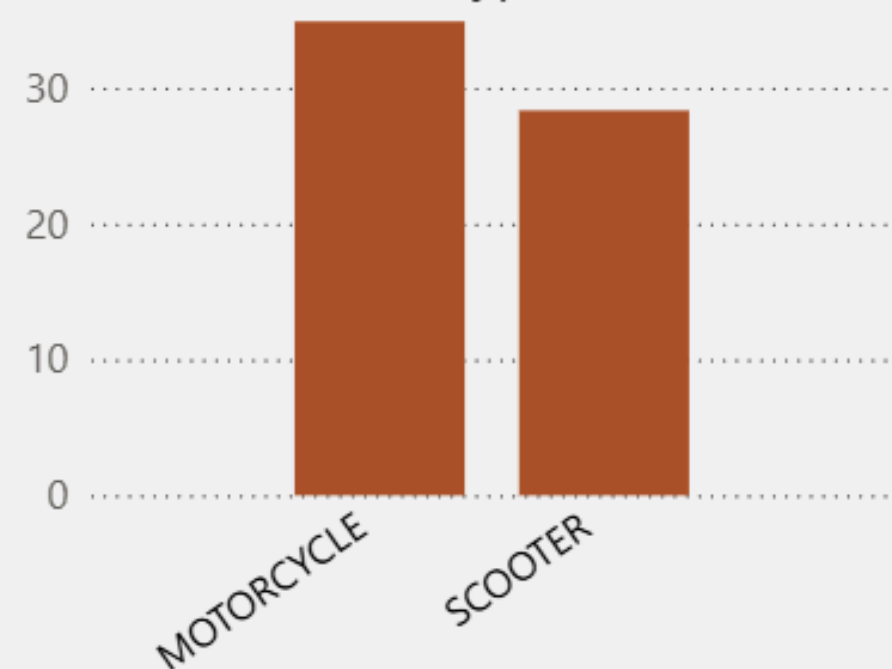
Average Delivery Time

33.25

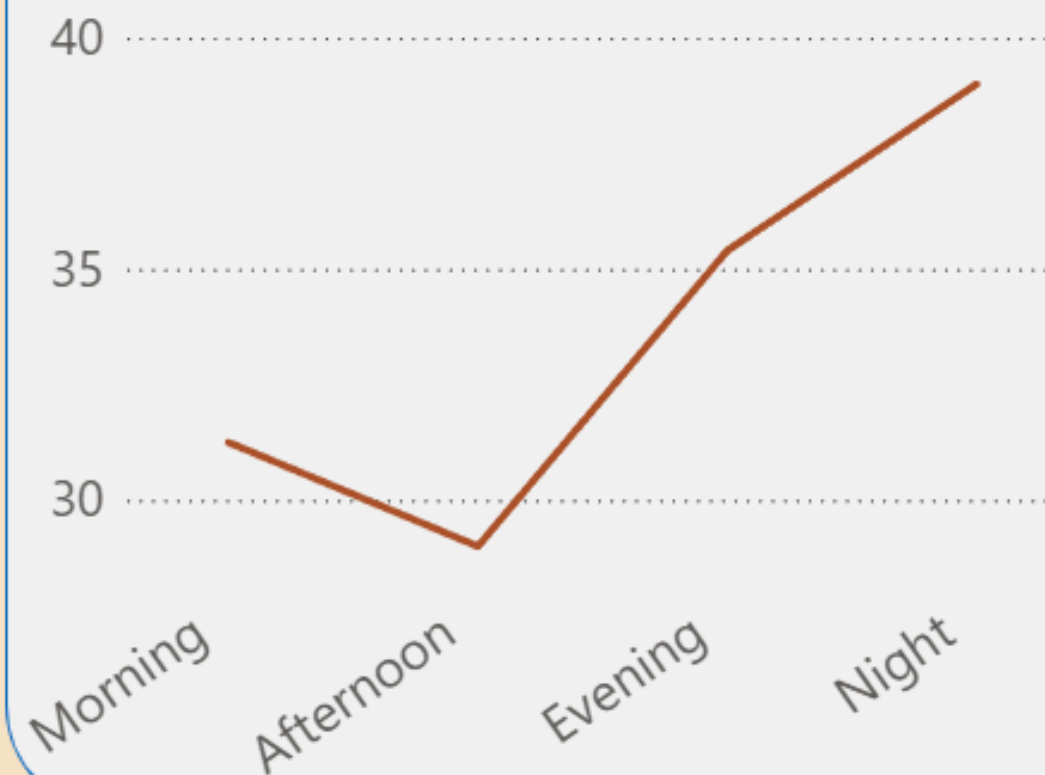
Max Delivery Time(Min)

49

Average Delivery Time by Vehicle Type



Average Delivery Time by Period



Min Delivery Time(Min)

24

Total Deliveries

12

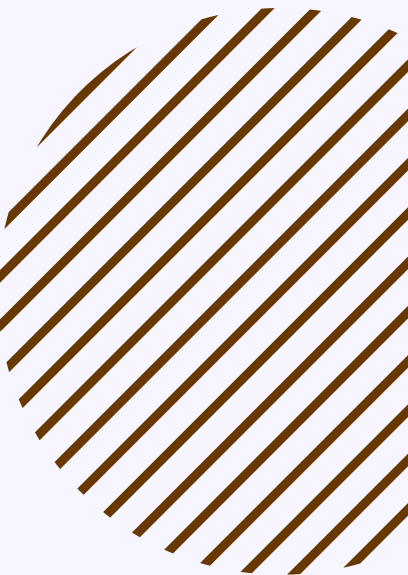
Key Insights & Business Impact

Key Insights:

- High Operational Scale: ~46,000 deliveries, on-time rate 70%
- Delivery Time Variability: 10–49 mins; inconsistent speeds reduce trust
- Traffic Bottleneck: High traffic = longer delivery times
- Weather & Festival Impact: Foggy/cloudy + festivals increase delays 20–30%
- Vehicle Type & Condition: Scooters/EVs fastest

Business Impact:

- Customer satisfaction and loyalty affected by inconsistent deliveries
- Operational costs increase due to delays and refunds
- Brand reputation suffers during peak/festival periods
- Growth without optimization amplifies inefficiencies



RECOMMENDATIONS

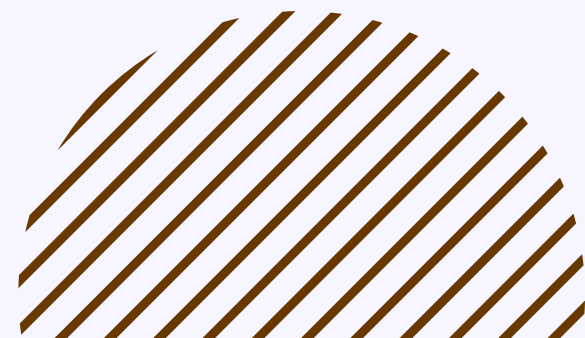
- Dynamic ETA & Dispatching
- Metro-Focused Capacity
- Time-of-Day Rebalancing
- Festival & Weather Planning
- Vehicle-Type Allocation
- Vehicle Health Monitoring
- Performance-Based Rider Development

LIMITATIONS

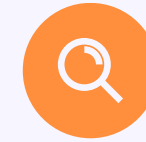
- No delivery distance or route complexity data
- Aggregated traffic categories (Low/Medium/High/Jam)
- Limited customer satisfaction integration
- Static historical snapshot (no seasonality trends)
- Small sample size for individual rider dashboards

FUTURE ENHANCEMENTS

- Integrate distance, routing, and GPS-level data
- Zone-level and city-level drill-downs
- Link delivery performance to ratings & cancellations
- Predictive models to forecast delays
- Track rider fatigue, shift length, workload patte



CONCLUSION



Zomato's delivery efficiency relies more on operational conditions than individual effort. Performance dips during traffic congestion, peak hours, festivals, and adverse weather. Implementing traffic-aware dispatching, vehicle optimization, time-based capacity planning, and rider performance management